

STRENGTH TRAINING TERMS

THERE ARE A plethora of different terms within strength training and it can be useful to know some of them in order to understand the effects of different training methods. We have tried not to use too many technical terms, but some are too important to pass by so we will explain them here.

STRENGTH TRAINING

We train muscle strength to generate more force. How much force a muscle can generate is dependent on:

- the size and number of individual muscle fibres. This is known as the muscular cross-sectional area.
- the degree to which we can activate each muscle fibre. Voluntary muscle contractions are caused by sending signals from the brain via the nervous system as electrical impulses. The number of impulses reaching the muscles over a given time interval will decide how much of the muscular cross-sectional area gets activated and hence the total force generated. The last 20 per cent of maximum muscle contraction is a direct result of this ability.
- anatomy. Where the muscle is attached via ligaments to the bones will affect the torque lever and therefore the actual force output, which can vary from individual to individual. A person with a longer torque lever will be able to output a higher force with less muscle mass than a person with more muscle mass but with a shorter torque lever.

Strength training is all about improving the first two qualities – namely muscle size and activation. Most climbers will benefit from focusing on getting stronger without getting bigger, and their training should, therefore, be targeted more towards improving their muscle fibre activation, and less on increasing the muscular cross-sectional area.

STATIC STRENGTH TRAINING

Static strength is when a muscle is generating power without creating motion.

The muscles related to finger and wrist strength are working mostly statically when gripping a hold, and sometimes we will do a move with an arm or a foot while the rest of our body is working statically to maintain the body position. We measure the static work by multiplying the hold time by the load.

When training fingers, the load is our body weight +/-added weight, and the hold time is usually 7 to 10 seconds for most methods of static finger strength training.

DYNAMIC STRENGTH TRAINING

Dynamic strength is when the musculature is generating force and thereby creating movement. This is the most common form of strength training, and in climbing the primary focus for dynamic strength training is to improve the strength of the pulling apparatus in the arms and upper body. The training load of dynamic strength training is measured in volume, which is the product of total weight lifted multiplied by the number of repetitions.

When doing pull-ups the training load equals your body weight +/- added weight multiplied by the number of repetitions multiplied by the number of sets.

CONCENTRIC

Muscles contract to create movement and to stabilise joints. A muscle contracting is in what is known as the lifting phase in strength training.

Lifting your body when doing a pull-up is the concentric phase of the movement.

ECCENTRIC

Using your muscles to slow down a movement is what we call eccentric training. When you're lowering yourself after doing a pull-up you're in the eccentric phase – also known as the breaking phase – of the movement. We will always be able to absorb more force than we can generate, and eccentric strength training can, therefore, be an effective method of strength training. It's important to remember though that this is a very stressful training method with a high risk of injury.

Eccentric strength training requires experience with strength training and a good training foundation to reduce the risk of injury. It will also lead to increased muscle soreness in the following days, which can negatively affect the quality of your other training.

PLYOMETRIC

Initiating a concentric phase by starting with an eccentric phase is known as plyometric strength training. Picture how you might dip low before exploding into a pull-up. Setting up the move like this exploits the elastic energy

created in the eccentric phase and enables us to develop more power in the concentric phase.

HYPERTROPHY TRAINING

The goal of hypertrophy training is to increase the muscular cross-sectional area. The size of a muscle has a significant impact on how much force it can generate, but in weight-dependent sports, like climbing, having overly large muscle volume is not advantageous. The way to increase muscle volume is through high training volume. This means training with a relatively low load per repetition, but increasing the number of repetitions to reach exhaustion for each set, or to train with a relatively high load and fewer repetitions per set. A normal dosage would be three to four sets of eight to twelve repetitions, with a resting period of 2 to 3 minutes between each set.

MAXIMUM STRENGTH TRAINING

The goal of maximum strength training is, as the name implies, to increase your maximum muscle strength. This type of training involves maximising the load for one to three repetitions. This will stimulate the activation of muscle fibres via the nervous system, more so than when doing hypertrophy training. For the last 20 per cent of a maximum muscle contraction, you're dependent on being able to activate all muscle fibres, and by prioritising this training method you'll be able to generate more force from the same muscular cross-sectional area, which, as a climber, is what you want. It's important to emphasise that the amount of force a muscle can generate is dependent on both the muscle size and the degree of activation, and most strength training programmes will, therefore, have a progression from hypertrophy training to maximum strength training.

Maximum strength training usually consists of three to five sets of three to five repetitions, with 3 minutes of rest between each set. More experienced athletes can train by doing four to six sets of one to three repetitions.

EXPLOSIVE STRENGTH

How much force a muscle can generate through a movement is dependent on how quickly it contracts. This is known as the rate of contraction. A muscle being strong is not the same as a muscle being able to generate force quickly, and it is essential for boulderers to train the ability of the muscles in the arms and upper body to contract quickly. Explosive strength training is characterised by relatively low loads – between 0 per cent and 60 per cent of max load – and with the fastest execution possible.

The most common regime is three to four sets of three to six repetitions, with a 3-minute rest between sets. Fast pull-ups, double dynos and big moves on the campus board are examples of climbing-specific explosive strength training.

RATE OF FORCE DEVELOPMENT

Rate of Force Development (RFD) is a measure of how quickly a muscle can go from zero to maximum force and is an important attribute of the muscles in the forearm for all kinds of climbing. This is an especially important attribute for bouldering and hard crux sections on routes, where you have to be able to quickly generate the force needed to hang on to small holds. We refer to this ability as contact strength. In climbing, we train this ability through maximum strength training like dead hangs, hard bouldering and campusing.

ENDURANCE

‘ENDURANCE IS THE ABILITY TO ENDURE.’

This quote is a simple yet very good definition of a complex phenomenon. Because herein lies the very essence: how long can you endure on a route or a boulder problem?

When we're talking about climbing endurance it's important to be familiar with the term 'pump'. Getting pumped is the feeling of getting tired in your forearms – and is often to blame when you fall off a route. There are two central components at play: the physiological and mental conditions. The physiological conditions include, among others, the number of blood vessels supplying oxygen to the muscular system of the forearms, the energy metabolism in the muscles, grip force, and the muscle's ability to recover during the release phase of a grip cycle. The mental factors affecting endurance are related to how tired we feel.

The mental aspect of all physical training, including strength and flexibility training, will affect our performance, and we will dive deeper into this in the chapter about mental performance factors. But when it comes to endurance training we feel that being familiar with the mental aspect is so important that we will present it here as a backdrop to the purely physical endurance training.



PHOTO: ANDY WICKSTROM

Maria Davies Sandbu gets a taste of the pump at Red River Gorge, USA.

MENTAL ASPECTS

The concept of the central governor model emerged as a response to what is known as an energy model to explain endurance performance. The energy model says, in short, that performance declines with the increasing degree of muscular fatigue throughout the activity, but what the model has trouble explaining is, for example, how the final kilometre of a 10-kilometre race is the fastest when world records are set. Or how we can be physically fit, but still end up getting pumped at the second bolt, long before the crux of our project. Because, strictly speaking, the pump is nothing but a sensation. Who hasn't approached the crux of a route, thinking only about how pumped their forearms are? Or wondered if they can clip the rope in time as their left hand crimp starts to open a little? Or experienced how quickly the pump seeps into their arms as they stand a few metres above the previous bolt, reluctant to fall?

Our brains are lazy and overprotective by nature and basically don't want us to get tired on a route. The feeling of being exhausted therefore arises significantly earlier than our capacity suggests and an important part of endurance training is to get to know this feeling. The thought patterns, 'Now I'm pumped, this will never go', and, 'Now I'm pumped, so I'll give it beans', will in all likelihood produce two different results, even though the physical conditions are the same. Being used to climbing with pumped forearms, knowing that you are able to still clip the quickdraws, daring to do a few more moves, knowing that you are able to stop and shake out some of the pump, that you are capable of doing hard moves even when pumped – these are all important mental attributes that constitute endurance performance. Being aware of these elements during training and actual attempts on a route will raise the quality of your climbing, and will eventually provide you with certainty as to what the pump means for you. And it is this confidence that will allow you to go all-in – or a muerte, as the Spanish say – on every single attempt.

An important element of the central governor model is also the expectation you have for the activity. For example, it is easier to endure climbing with pumped forearms if you know you are coming to a good rest in three moves, versus on an on-sight when you don't know what's in store after the next three moves. Of course, there is a lot of training behind managing expectations, whether it is the result of working on a route, general endurance training or good mental preparation. And it's exactly these preparations that are so important for your expectations, both the physical and the mental. For it is beyond any doubt that physical attributes are crucial to endurance performance, and these are trained through physical conditioning. The mental preparations lie in coping with climbing while pumped, visualising how the sequences on the route are best solved and how they will feel, and finding the correct level of arousal. These are elements we will discuss in more detail in the chapter on mental performance factors.



PHOTO: NATHAN WELTON

Rannveig Aamodt on-sighting in Red River Gorge, USA.

PHYSICAL ASPECTS

The mental aspects of endurance are gaining increasing attention, but there is a broad consensus that we must be physically fit enough to produce good endurance performances.

When doing repetitive movements, such as gripping and letting go when climbing, the metabolism in the muscles determines how many repetitions we can manage, and this can be simplified into two forms: with and without an adequate supply of oxygen. This is called aerobic and anaerobic metabolism, respectively. Oxygen is transported from the lungs via the blood to the muscles at work. For several traditional endurance sports, such as running and cross-country skiing, the circulation system, oxygen uptake and utilisation rate are considered important factors. Metabolism is primarily aerobic when working at lower intensities and with a well-developed circulation system. When working harder and at higher intensities, anaerobic metabolism is required to maintain the muscular work. Degradation of nutrients such as glycogen and creatine phosphate in the muscles is then the primary source of energy, a by-product of which is lactate, also known as lactic acid. The accumulation of lactate that we are unable to dispose of in the muscles is considered a decisive factor in our reduction of intensity, and will also have consequences for both fine coordination and how quickly the muscles can contract. This has obvious consequences for our climbing performance, as both being able to coordinate a move and being able to quickly close the grip on the next hold is crucial to whether we fall or not.

Climbing is not, in principle, a sport that places any special demands on oxygen uptake or the circulatory system. It is nevertheless difficult to argue against it being advantageous to be in generally good physical shape given the capacity we have to endure training and how quickly we recover. Still, we won't improve our climbing-specific endurance by only running intervals. Climbing places significantly greater demands on the local conditions of the forearm muscles, including the number of blood vessels in the muscles, the ability of the muscles for anaerobic metabolism, and maximum finger strength.

The number of blood vessels in the muscles is important for transporting oxygen to the forearm muscles. As we grip harder, we will increasingly shut off blood circulation in the forearms. It is estimated that with a grip load between 40 and 75 per cent of maximum finger strength, we have completely shut off the blood supply. At lower loads, however, it's still possible for blood to circulate to the forearm muscles, and the number of blood vessels will then be crucial for nourishing the working muscles. Even more important, however, is the blood circulation in the muscles as we release the grip. This removes the by-products from anaerobic metabolism and ensures that we recover more efficiently between each gripping phase. Exactly this trait appears to be crucial for endurance performance in climbing. Since a climbing route will involve repeated grip and release phases, our ability to recover in the release phases is important for how many moves we can do.

Good maximum finger strength gives us a better margin per grip and allows the blood to circulate also during the grip phase, allowing the muscles to last longer before they tire. This is even more important on more intense routes that leave little room for rest in the release phases, as a release phase shorter than three seconds is not enough for the muscles to recover.



PHOTO: CHRIS BURKHARD

Stian Christophersen climbing into the sunset in Kalymnos, Greece.

Since routes of this type will provide inadequate muscle recovery, the muscles' ability for anaerobic metabolism will be crucial. There are two simple methods for testing this ability:

1. Use an edge you can hang on to for at least 15 seconds. Hang in intervals of 10 seconds, with a 3-second rest between each interval, for as long as you can. This will give you an indication of how well you can recover during the release phase. The edge depth provides an indirect measure of your finger strength.
2. Use an edge you can hang on to for at least 15 seconds. Hang until exhaustion. This will give you an indication of how well your muscles are working anaerobically and an indirect measure of your finger strength.

It is important to distinguish these two traits in training as they are trained in different ways. The ability to recover in the release phase and increasing the number of blood vessels in the muscles are trained at a lower intensity for a longer duration, while how well the muscle works anaerobically is trained at high to maximum intensity and a duration of 30 to 120 seconds. These two training methods train your endurance and your power endurance, respectively. To determine the intensity of the workout, we use a five-step scale from no pump to maximum pump, and the different methods can be divided according to this scale:

ENDURANCE	INTENSITY LEVEL 1-5	POWER ENDURANCE	
Arc and continuous	1	3-5	Circuits
Circles	1-3	4-5	4 x 4 boulders

Yo-yo	2–3	4–5	The McClure Method
3 x 5 routes	2–3	4–5	Deadhangs
2 x 4 routes	3	4–5	1 x 4 routes

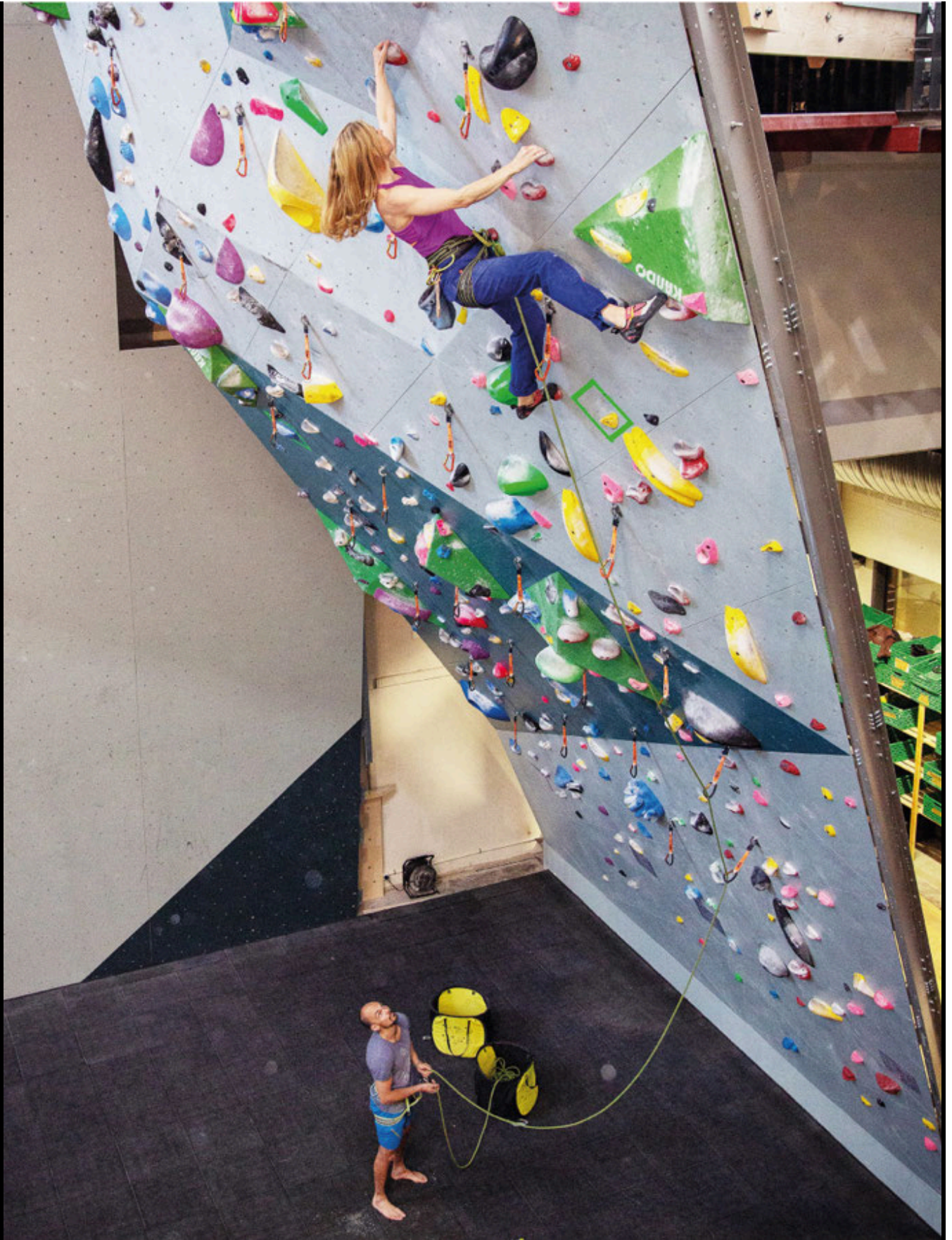
LEVEL 1: No pump, endurance

LEVEL 2: Slight pump, endurance

LEVEL 3: Moderately pumped, endurance

LEVEL 4: Very pumped, power endurance

LEVEL 5: Extremely pumped, power endurance



ENDURANCE TRAINING

By training endurance, we will increase the number of routes we can do in one day or in one session, and how many easier moves we can do in a row without getting drained. The training helps us preserve our strength during longer, easier sections of a route and helps us recover during a climb. Because endurance training also gives us a high volume of moves, it will also build a foundation to endure more and harder training in the future. All together, we can say that endurance training enhances our capacity, both on the wall and in our training.

Common to these training methods is that they have a high volume – the number of moves or the number of minutes on the wall – and low intensity. Here are six methods for training endurance.

1. ARC (AEROBIC ENERGY RESTORATION AND CAPILLARISATION)

This method is comparable to mellow long-distance running. The advantage of this method is that it builds your capacity to endure training and to recover on the wall during a climb, but the disadvantage is that it takes both a long time and is at a level so far below your own maximum level that it might not be very technically demanding. The most rewarding way to train this, we think, is to do lots of easier routes outdoors, for example as an active rest day, as this provides a greater benefit for your technique than a corresponding indoor session.

2. CONTINUOUS

This training method is aimed at the formation of new blood vessels in the muscles and to build a base for more intensive training. It involves being on the wall continuously for over 10 minutes at a sufficiently low intensity so that you do not exceed level 2 on the 'pump scale'. As your endurance progresses you can continue climbing beyond the initial 10 minutes, but one repetition is enough per session.



3. CIRCLES

Circle training is a popular method based on interval principles. This means periods of work and periods of rest where the intensity per circle, the rest time between the circles and the total number of circles determine how hard the training will be. When endurance training at lower intensities (levels 2 to 3), each circle can be between 4 and 10 minutes long, and involve trying not to get pumped while climbing. Here you can easily insert some harder sections followed by easier sections and a rest so that you get variation through the circle and allow the pump to come and go, a bit like when you're running in hilly terrain. You can start with two circles per session with a 2- to 3-minute break between each circle and increase both the number of circles and the time per circle as your endurance improves. The easiest approach is to use a training wall with lots of holds and use any handholds and footholds, but you can also get your training partner to point out which holds to use as you climb.



4. YO-YO

Climb a route on lead, then down-climb the same route, and then climb back up a second route on top rope. The total time should be 5 to 7 minutes on the wall per set. The difficulty level per route and the number of sets in total determines how hard the session is, but for endurance training, the focus should be on easier routes and more sets. Yo-yo sessions are easier with a climbing partner who is doing the same session as you, so that the rest period is roughly equal to the climbing time. Begin training with two to three sets and increase to five sets before making each route a little harder.



5. 2 x 4

Choose two routes where the first one is a little harder than the second and lead them one after the other. Done correctly, you should be able to complete the first route by a small margin, with a moderate forearm pump, and you should be able to recover on route two. Repeat in sets of four with a rest period equal to climbing time. If possible, it is good to vary the routes in the sets in order to maximise the variation in your climbing.

6. 3 x 5

Choose three routes that are roughly equally hard and lead them one after the other. Repeat five sets, and adjust the resting period to match the climbing time. If possible, it is good to vary the routes in the sets in order to maximise the variation in your climbing. The goal is to keep the pump at an acceptable level so that you can complete the exercise with only a light to moderate pump. A common mistake is to start too hard; our recommendation is to choose routes that are a tad easier than you think will allow you to complete the session.



You will climb faster on circles that you're familiar with, so make your own circles longer than when your partner is pointing them out while you're climbing.



Vary the length of the circles, both during a session and between each session.

POWER ENDURANCE TRAINING

Power endurance is a key quality for managing hard routes. While we have the opportunity for longer release phases and even rests on easier moves and sections on a route, harder sections – or short and hard routes – do not allow for this in the same way. As mentioned earlier, the muscles will not be able to recover during hard climbing and must work with an ever-increasing accumulation of lactate and other by-products from anaerobic metabolism that occurs during the grip phases. This gives us an increasing degree of pump and muscular fatigue which not only weakens the muscles physiologically but also weakens our mental ability to maintain intensity. This demands that we train not only the physiological properties to cope with this, but also our mental ability to climb with pumped forearms. Learning to climb in a technically correct way and making good tactical choices on a route while the forearms are burning is just as important as training for physiological adaptations in the muscles. The basic principle of power endurance training is therefore that it is performed with high intensity: high up on the pump scale and with a lower volume, which means fewer moves or minutes on the wall.

1. CIRCLES

Here are two methods for training power endurance using circles:

- POINT IT OUT** . Have your climbing partner or coach point out holds for you. This requires being familiar with the wall, the holds on the wall and the level of the climber so that you reach the required number of moves or minutes on the wall. If the moves are too hard, each circle will be broken up by too many falls and short sections, while if the moves are too easy you won't achieve a high enough intensity. The climb should last for about 90 to 180 seconds, which is somewhere between 25 and 50 moves, depending on how fast you climb. You should do four to six circles per session with 3 to 5 minutes of rest between each circle.

- REDPOINT** . Here you can create your own route with defined handholds and footholds. The climbing time should be around 90 to 180 seconds, or 40 to 60 moves. You can also simulate the type of climbing that you are training for. So, if you're training for a particular route, you can create a circle that simulates both the number of moves and how the sections are put together. For example, if there are three crux sections with rests in between, your circle can reflect this. If the route is more homogeneous with little or no rest, the circle can be created to simulate this. Making these circles yourself is also great training for your technique since you have to test and adjust the moves, handholds, and footholds several times to get it to work. You can vary the sessions by having two to four attempts on very hard circles with a long (over 10 minutes) rest between each attempt, and having slightly easier circles, resting for 3 to 5 minutes with a total of four to six attempts.



2. 4 x 4 BOULDERING

Choose four different boulder problems which force you to try hard, yet are still relatively manageable. Each problem is repeated four times in a row without rest, which amounts to one set. Then rest for 3 minutes before repeating this for the next three problems. We recommend choosing relatively sustained problems with between 8 and 12 moves. The problems should be on overhanging walls and not too technically demanding.



3. THE MCCLURE METHOD

This method gets its name from British climbing legend Steve McClure, who uses it as one of his exercises to prepare for hard and intense redpoints and on-sights. The exercise is done on the campus board with the legs either on a chair or a foothold, and you go up and down continuously – for example, 1-2-1 or 1-2-3-2-1 – for 60 to 120 seconds. If you manage more than 120 seconds, adjust the foot position to make it more difficult. This is then repeated for five to six sets with a 2- to 3-minute break between each set.

4. 1 x 4 ROUTES

Here you choose four different routes which are so hard that there is a high chance that you will fall at the top. They should also be as sustained as possible. You should climb until you fall, and the goal is to go all-in on every attempt. Rest for 10 minutes and move on to the next route. If there aren't enough routes at your climbing wall you can have four attempts on the same route.

5. DEADHANGS

Deadhanging is a simple and effective high-intensity endurance training method because we can customise how long the grip and release phases are for each set.

To make this training specific enough, we need to look at how long the grip and release phases are when climbing.

Observational studies have shown that the average grip phase in competition lead climbing lasts for about 7 seconds. As mentioned earlier, the release phase must last for more than 3 seconds in order for us to recover during a climb. When we put this together for a deadhang session, it can look something like this:

Hang time	Rest between reps	Repetitions	Sets	Rest between sets
7 seconds	3 seconds	7	7 to 14	2 minutes

Hanging and resting times are adapted to the grip and release phase during climbing. The goal of the session is to train each set to exhaustion, to the point where the fingers start to open. In order to have progression and variation in your training, you should use different grip positions and edge depths.

Start with one session per week and increase to two sessions with a minimum of two days of rest between each session after four to six weeks of exercise. Since these sessions are both load and volume intensive, it is recommended that you don't combine them with regular climbing in the same session.



Beastmaker is a fingerboard developed by British climbers. It is available in two models, the Beastmaker 1000, which has the better holds, and the Beastmaker 2000. Each model comes with its own training programme, with different levels of difficulty, which can be accessed through the Beastmaker smartphone app. These programmes are based on repetitions and sets, as described in the model above, and ensure a variation of grip positions throughout the programme. Overall, these pre-made programmes constitute many hangs in different grip positions and can, therefore, be used as a good alternative for finger training when you just don't have the time to go to the climbing wall. Although you will get stronger fingers, these programmes can be considered a sort of hybrid between strength and endurance training which will cover a wider range than just maximum strength training for your fingers.



Our experience with the app is that the exercises described are often hard to complete according to the climbing grade they are associated with. Therefore, go down one grade compared to what you normally climb, and complete the entire programme before considering whether to increase the level of difficulty. These programmes train you to exhaustion, and the goal of each session is to complete all the hangs as prescribed. When you find that you can complete a session with a fair margin, you can try to go up one level.



CHICKEN WINGS

Have you wondered why we climbers raise our elbows when we're getting pumped? This strange phenomenon we call chicken winging is caused by us automatically trying to find a better angle on the handholds for the flexor muscles of the fingers. As the muscles tire when climbing, the elbows start to rise and bend backward in relation to the wrists, and it looks like we have chicken wings.

So now you can at least see when your partner is about to fall and give him the scientific explanation for his chicken wings when you've lowered him to the ground – though that's probably the last thing he'll want to hear about at that point ... !



DIFFERING PERFORMANCE FACTORS BETWEEN BOULDERERS AND LEAD CLIMBERS

THERE IS STILL a lot we do not know about the physical differences between boulderers and lead climbers, but some elements are beginning to emerge as important.

First and foremost, we must look at the requirements posed by the different styles of climbing. Bouldering places greater demands than lead climbing on finger, arm and upper body strength due to harder single moves, poorer holds and fewer moves. In addition, bouldering demands rapid movement and stopping of the centre of gravity to avoid swings. These two requirements are reflected in the fact that boulderers have better maximum finger strength and faster force development in the forearm muscles than lead climbers. We can assume that the same traits are important for the arm and upper body musculature, and training for those who want to specialise in bouldering must, therefore, be adapted to encourage these traits. With an increased demand for force development, muscle mass will also often follow suit, which may explain why boulderers appear to be somewhat heavier than lead climbers.

Although lead climbers also depend on strong fingers, greater demands will be placed on endurance factors, such as the ability to recover during the release phase and how long they can climb before the muscles are exhausted. Lead climbers generally score better on these parameters than boulderers, which in turn indicates that these are attributes that you must emphasise if you wish to specialise in lead climbing.

In competition climbing there has been a clear distinction between lead and bouldering during the last decade, where the boulder problems were set with moves rarely or never seen in lead competitions, while the lead competitions often had routes with up to 60 or 70 moves. This distinction made it extremely difficult to combine the two styles, and there have been only a handful of climbers worldwide who have managed to successfully combine the two disciplines internationally in recent years. Nevertheless, there is now a tendency for lead climbing to incorporate more of the dynamic elements from bouldering, and the fact that the time limit – the time one is allowed to use on a route – has decreased from 8 to 6 minutes has meant that the number of moves per route has gone down. Bouldering, for its part, has retained much of its distinctive style of route setting, but it now seems more feasible for competition climbers to combine the two styles, which is clearly an incentive from the International Federation of Sport Climbing to make lead climbing more exciting and spectacular to watch.

For those of us who do not compete at a high level, we need to look at the demands of the different styles of climbing, and thus what factors are most important if we want to specialise or train specifically for bouldering or lead for a period of time. In short, physical training for bouldering should emphasise to a greater extent maximum and explosive strength in fingers, arms and upper body, while physical training for lead climbing should focus more on the endurance elements presented earlier in the chapter. Still, as highlighted several times, climbing performance depends on far more than the physical, and there is a lot to be learned from both styles of climbing, even if you prefer one over the other.



PHOTO: MIKE HUTTON

Maddy Cope on Snake Charmer (E5 6c), Huntsman's Leap, Pembroke, Wales.



MOBILITY

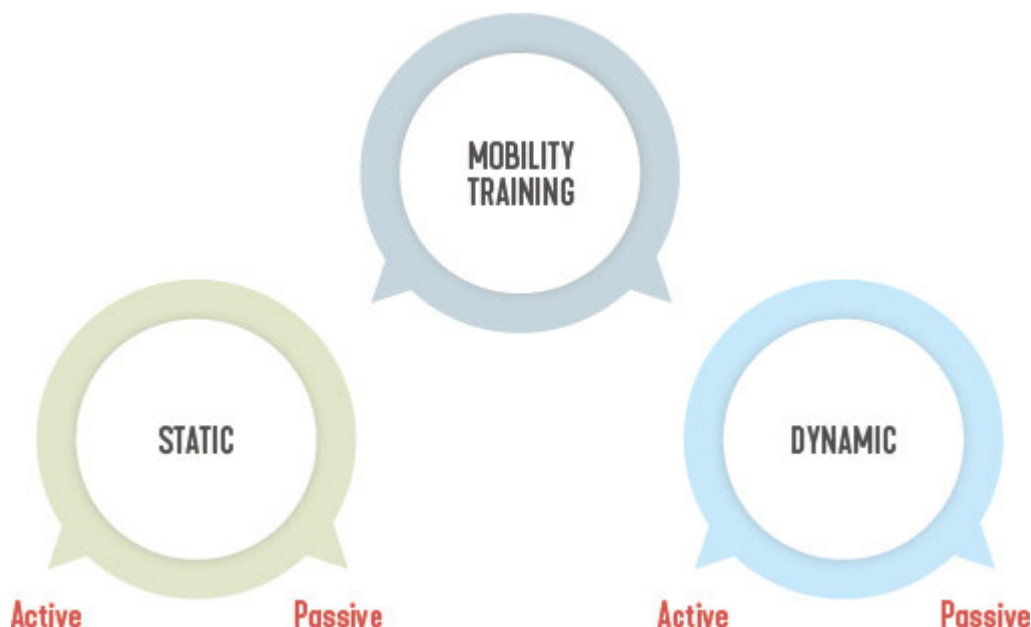
Our mobility determines the range of motion of our joints and thus has an impact on how we solve various climbing moves. Imagine having to place your foot high to do a rockover, or reaching your foot high out to one side in a corner. These are moves that require mobility of the hip joint in several different directions. Many climbers compensate for a lack of mobility by solving moves in different ways, but this often requires more effort than is necessary. Having the mobility to be able to perform specific moves, as well as solving moves in as energy efficient a way as possible, are the main reasons why we believe it is important to do mobility training.

There are a number of factors that affect our mobility, including the anatomy of the joints, the mechanical properties of connective tissue, muscles, tendons and ligaments, and neurological factors. Mobility training methods are methods to influence these structures so that the range of motion increases.

We will now look at what training methods we can use for mobility training and which joints we think are appropriate to focus on to improve climbing performance and potentially reduce the risk of injury.



MOBILITY TRAINING CAN BE DIVIDED IN THE FOLLOWING WAY



Static mobility training means pushing the joints slowly towards full range of motion either by applying your own muscular force (actively) or by using external forces such as your own body weight or a partner (passively). When you reach the point where you're no longer able to extend further, hold this position for the given time interval. This is traditionally the most common training method for mobility. It is easy to learn and implement and is proven to have a positive effect on range of motion. It is important to take into account the intensity of the stretching, as excessive force can cause damage to the muscles, joints and connective tissues. Another important issue is that static mobility training has a negative effect on the muscle's ability to produce force and power immediately after stretching, and we recommend that you do not do these kinds of exercises immediately before climbing.

Dynamic mobility training means pushing the joints to full extension several times in succession, but without holding this position over time as you do with static mobility training. It is most common to do this actively, i.e. by using your own muscle power. This can be done as a separate session in itself, or form part of a warm-up programme for climbing, as this kind of exercise has not shown the same immediate negative effects on the musculature as static mobility training has.

Below is an example programme for both static and dynamic mobility training:

METHOD	EXECUTION	FORCE	HOLD TIME	REPETITIONS	REST	TIME PER SESSION	SESSIONS PER WEEK
Static	Stretch slowly until end of range and hold position	High	20–60 seconds	2–6 per exercise	30–60 seconds	15–60 minutes	2–4
Dynamic	Repeatedly stretch until end of range and back with increasing speed	High	N/A	Over 6 per exercise	30–60 seconds	15–60 min	2–5

Mobility training to the extent described above is best done in a session on its own, but incorporating some exercises into your warm-up routine before climbing may also be appropriate. This will increase the blood circulation to and elasticity of the tissue and may help you move better as you climb. Just be aware of static stretching with a long hold time just before climbing, because, as we mentioned, this will lead to reduced muscle strength for a period of time.

As we see it, mobility in the hips, shoulders and upper back is especially important for climbing, and we will now take a closer look at mobility training for each of these areas.



HIPS

Good mobility in the hip flexor means that we can lift the knee, and thus the foot, high, and this is important for high foot placements. Lack of mobility here means that we either have to pull ourselves higher with our arms to get our feet up, or move our hips further out from the wall. Both of these options cost a lot of effort when the holds are bad, as they will become even worse when we pull ourselves higher and our centre of gravity moves out from the wall.

When fronting, we rotate the hip out so that the knee points away from the centre of the body and the inside of the thigh faces the wall. In this way, the body is held closer to the wall and the centre of gravity can be positioned directly above the foot so that the upward movement is created more by the leg than by the upper body.



In addition, it is important to be able to extend the leg straight to the side, both to reach faraway footholds, and to be able to apply pressure and gain balance with a wide platform between two footholds which are far apart.

Good mobility in this movement will, like with the hip flexor, allow you to lift your leg further to the side without using more power in the upper body than necessary.

STATIC STRETCHING

1. LOTUS

Sit on the floor with the soles of your feet facing each other and your knees pointing out to the side. Straighten your back and push your knees down to the ground. You can push your knees down with your arms, or have a partner help you. You should feel the inside of your thighs stretching up towards the groin. Hold this position for 40 to 60 seconds, release for 15 seconds and repeat two to six times, gradually stretching more every time.



Your back should be straight for all of the following exercises!



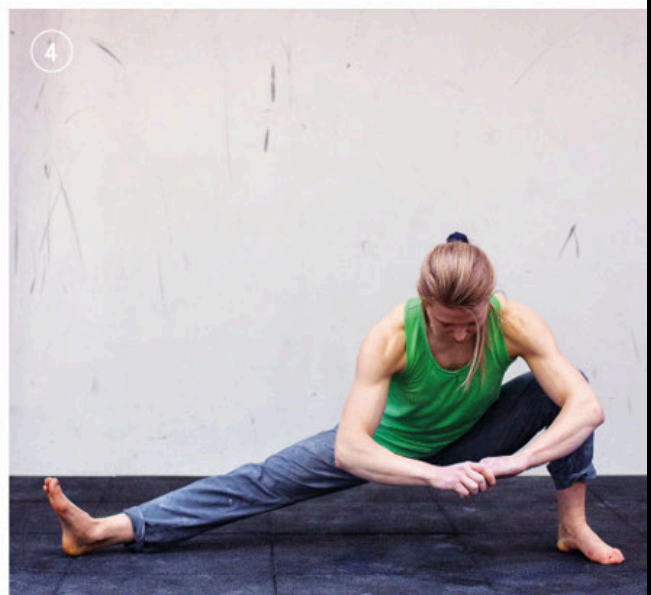
2. SPLITS

VERSION 1: Sit on the ground with your legs pointing to the side. Lie forward as if trying to push your chest down towards the floor. You should feel the stretch behind and inside the thighs. Hold the position for 40 to 60 seconds, release for 15 seconds and repeat two to six times with gradually increasing force.



VERSION 2: Stand upright, slide your feet out to each side and lower your hips towards the floor. You should feel the stretch on the inside of the thighs. Hold the position for 40 to 60 seconds, release for 15 seconds and repeat two to six times, gradually stretching more every time.

DYNAMIC STRETCHING



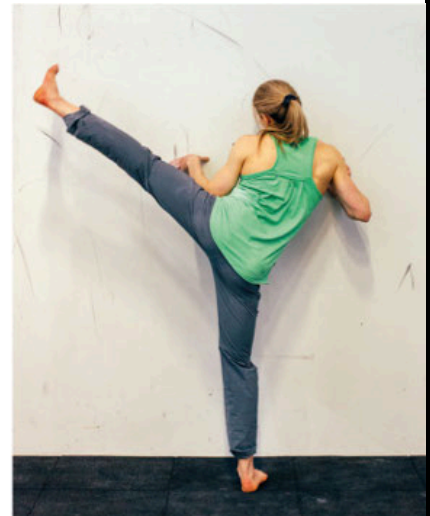
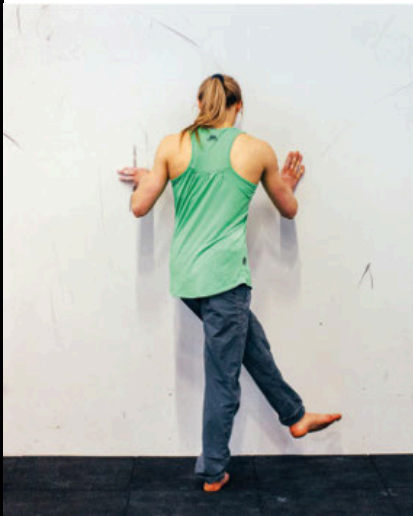
1. COSSACK SQUAT

Stand upright with your legs out to the side. Shift your weight towards the right and squat as deep as you can with most of your weight on the right leg. Your weight should be resting on the right heel so that the sole of your foot is in contact with the ground; you should feel it stretching the groin and the back of the thigh. Stand back up into the starting position before shifting your weight over the left leg and squatting to the left. Repeat for six sets, with four squats per leg, and hold the squat position for 10 seconds while gradually stretching more every time.



2. OPEN SQUAT

Stand up against a wall with your feet pointing out to each side, with your knees and thighs as close to the wall as possible. Lower your hips until you feel the stretch on the inside of your thighs and up towards the groin. Hold this position for 5 to 10 seconds before standing back up. Repeat three to four sets of eight to ten repetitions, gradually stretching more every time.



3. LEG SWING

Stand upright facing the wall, using your arms against the wall as support. Cross your left leg in front of your right and then quickly swing it back left and up so that your toes are pointing up towards the ceiling. You should feel it stretching the back and inside of your left thigh. Repeat three to four sets of 20 repetitions for each leg, gradually stretching more every time.

SHOULDERS AND UPPER BACK



Lie down on your back and let your arms drop backwards to the floor. Are you able to touch the floor with both your hands without arching your lower back?

Shoulder mobility is important in climbing, and the shoulder is also dependent on the mobility of your thoracic spine – the upper part of your back where the ribs are attached to your spine – to function properly.

Good shoulder and thoracic mobility will allow us to reach the next hold without compensating by moving the upper body too much. This allows us to keep our body more stable when doing a move. In addition, all variations of stemming between holds and rotations for large crossover moves require good shoulder and thoracic mobility. Mobility training for the shoulders must therefore also include thoracic mobility training. We have chosen to focus on active, dynamic exercises in order to work with multiple joints at the same time.



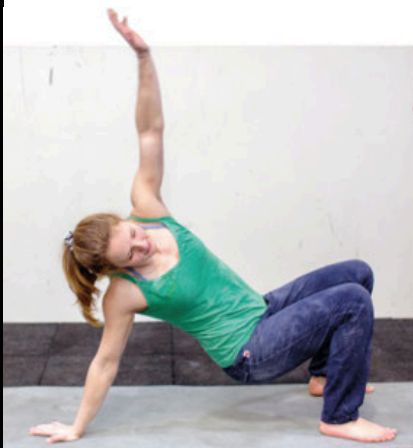
1. KNEELING ARCH

Kneel on both knees and place your hands on your heels for support. Push your hips forwards and your shoulders back so that the insides of your arms point forwards and your chest points up towards the ceiling. Hold this position for 3 to 4 seconds before relaxing. Repeat two to three sets of 10 to 15 repetitions. You should feel it stretching the front of your hips, chest and shoulders. The exercise becomes harder the closer together your hands are, and this is adjusted by how wide you place your feet.



2. MOBILITY ON A STICK

Grab a stick with both hands. Move your arms over your head and down your backside, then back up and over again. You should feel it stretching across your chest, and the exercise is harder the closer together you grip with your hands. Do the exercise slowly, relaxed and in control all through the movement. Repeat two to three sets of 15 to 20 repetitions.



3. THORACIC ARCH

Start with your right hand on the ground behind you and both feet pointing forwards. Push your hips and chest up towards the ceiling and get up on your toes. Move your left hand behind you and to the left as if you're about to place it on the ground behind you on your left side. Hold this position for 3 to 5 seconds before

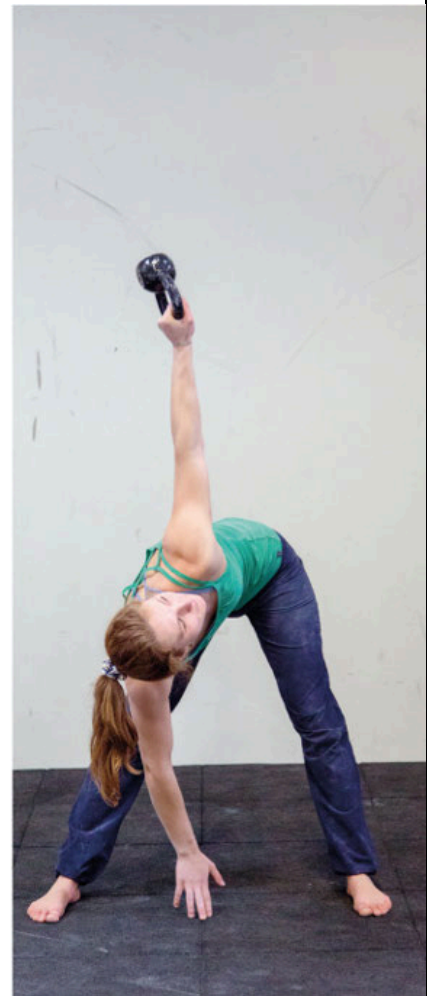
returning to the starting position. Repeat two to three sets of eight to ten repetitions for each arm. You should feel it stretching the front of your hips, chest and shoulders.



4. ROTATING STARFISH

Start as if doing a push-up, and rotate up into a sideways starfish position by lifting your right arm and leg. Continue the rotation by placing your right hand and leg behind you and then lifting your left arm and leg as

you continue around and back to the starting position. Complete four full rotations to each side, eight in total, for two to three sets. The exercise can be made easier by not lifting your leg as you rotate.



5. ROTATING SHOULDER PRESS

Hold a dumbbell or a kettlebell in your left hand and lift it up until your arm is straight. Keep your arm straight as you bend forwards and rotate your upper body until you can touch the floor with your right hand. Hold this position for 3 to 5 seconds and return to the starting position. Repeat for two to three sets of eight to ten repetitions. The exercise can be made more difficult by standing with your feet closer together, and harder by using heavier weights. Holding a kettlebell upside down will make it more difficult to stabilise your arm and can add an extra challenge to the exercise. You should feel the stretch in the back of your thighs and you should be actively engaging your upper arm and the muscles between your shoulders to keep the arm straight all through the movement.

REMEMBER

It's important to maintain body control during all these exercises. Even when doing a dynamic exercise like the leg swing you should feel in control of the movement and gradually increase the range of movement until

you've reached the maximum range of motion of the joint. It's important to control your breathing and relax as much as possible. If you're pushing too hard and you hold your breath, it'll only be harder for you to reach full range of movement. Stretching too fast or with too much force will also increase the risk of injury to your muscle tissue, so it's important to start slowly and gradually increase the speed and how hard you're stretching as you get more familiar with the exercises.

* Failure also means technical failure. If you lose form in your hang posture or can't maintain your finger position, you've reached technical failure and should step off.

1 López-Rivera, E. and González-Badillo, J.J. (2012). The effects of two maximum grip strength training methods using the same effort duration and different edge depth on grip endurance in elite climbers. Sport Technology 5: 1–11. López-Rivera, E., and González-Badillo, J.J. (2016). Comparison of the Effects of Three Hangboard Training Programs on Maximal Finger Strength in Rock Climbers. In Northern Michigan University (Ed.), 3rd International Rock Climbing Research Congress, Telluride, USA 5th-7th August 2016. Telluride (Colorado, USA): IRCRA.

2 <http://en-eva-lopez.blog-spot.com.es/search/label/Finger%20training>

3 [evalopeztraining](#)
